PROJECT REPORT

On

Ai Carrier Guidance System

Submitted in Partial Fulfillment of Award of

BACHELOR OF TECHNOLOGY

In

Computer Science and Engineering

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ALLIANCE SCHOOL OF ADVANCED COMPUTING

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CERTIFICATE

This is to certify that the project work entitled Ai Carrier Guidence System submitted by Kota Sai Kruthic[2411021060154],Gnaneshwar[24110210175], Bharth Kumar[24110210177]and Rishikesh [24110210116]partial fulfillment for the award of the degree of Bachelor of Technology Computer Science and Engineeringof Alliance University, is a bonafide work accomplished under our supervision and guidance during the academic year 2025-2026. This thesis report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

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DECLARATION

We hereby declare that the project entitled **Ai Carrer Guidance System** submitted by us in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (CSE) of ASAC, Alliance University, is a record of my/our work carried under the supervision and guidance of Dr R C Karpagalakshmi.

We confirm that this report truly represents the work undertaken as a part of our project work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the faculty guide.

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TABLE OF CONTENTS	Page Number
1. INTRODUCTION	6
2. LITERATURE SURVEY	7
3. SYSTEM DESIGN	8-9
4. SYSTEM IMPLEMENTATION	10-11
5. RESULTS AND DISCUSSION	11-12
6. TESTING	12
7. CONCLUSION AND FUTURE ENHANCEMENTS	12-13
8. RFERENCES & BIBLIOGRAPHY	14

INTRODUCTION

In the modern era, career selection has become a complex and critical decision for students and young professionals. With the rapid growth of industries, emerging technologies, and a vast number of career options, individuals often struggle to identify the most suitable path based on their abilities and interests. Many students rely on peer influence, societal pressure, or limited traditional counselling, which may lead to incorrect career choices and dissatisfaction in the long run.

An **AI Career Guidance System** is an advanced, technology-driven solution designed to address these challenges by providing intelligent and personalized career recommendations. This system leverages the power of **Artificial Intelligence (AI)**, **Machine Learning (ML)**, and data analytics to evaluate an individual's profile and suggest the most appropriate career paths.

The system typically collects input data such as academic performance, skill sets, interests, personality traits, and sometimes aptitude test results. Using trained machine learning models and decision-making algorithms, it analyzes this data to identify patterns and correlations between user profiles and successful career outcomes. Based on this analysis, the system generates tailored recommendations for careers, courses, and required skill development.

One of the key advantages of the AI Career Guidance System is its ability to continuously learn and improve. As more data is collected, the system becomes more accurate in predicting suitable career paths. It can also integrate real-time labor market trends, industry demands, and future job forecasts, ensuring that users receive up-to-date and relevant guidance.

Additionally, the system can provide detailed insights into each recommended career, including:

- Required qualifications and certifications
- Skills needed for success
- Job roles and responsibilities
- Salary expectations and growth opportunities
- Future scope and industry demand

Literature Survey

The use of **Artificial Intelligence (AI)** in career guidance has gained significant attention in recent years due to its ability to provide personalized and data-driven recommendations. Traditional career counseling methods often lack flexibility and personalization, which can lead to ineffective career decisions. To overcome these limitations, researchers have explored AI-based systems that can analyze individual profiles and suggest suitable career paths.

Several studies have shown that AI-powered career guidance systems use techniques such as **Machine Learning (ML), Natural Language Processing (NLP), and data analytics** to evaluate user inputs like academic performance, interests, skills, and personality traits. These systems compare user data with trained datasets to identify patterns and recommend appropriate career options.

Research has also focused on the use of different machine learning algorithms, including **Decision Trees, Random Forest, Support Vector Machines, and K-Nearest Neighbors (KNN)**, to improve the accuracy of career predictions. These algorithms help in classifying user profiles and matching them with relevant career opportunities based on historical data.

In addition, some studies have introduced **deep learning models** such as Long Short-Term Memory (LSTM) networks to enhance prediction capabilities. These models consider multiple factors, including academic records, extracurricular activities, and personal goals, providing more comprehensive and accurate career recommendations.

Another important area of research is the development of **chatbot-based career guidance systems**. These systems allow users to interact in a conversational manner, making the guidance process more user-friendly and accessible. Chatbots can collect user information efficiently and provide instant career suggestions.

Personality-based career guidance systems have also been explored, where psychological models like the

Big Five personality traits are used to assess an individual's behavior and preferences. This approach ensures that the suggested careers align not only with skills but also with personality, leading to better career satisfaction.

SYSTEM DESIGN

The **AI Career Guidance System** is designed to provide personalized career recommendations by analyzing user data using intelligent algorithms. The system follows a structured architecture consisting of multiple components that work together to collect data, process it, and generate accurate career suggestions.

User → User Interface → Data Processing → AI/ML Model → Recommendation Engine → Database → Output

- **User** → Enters details like marks, skills, interests
- **User Interface** → Takes input (website/app/chatbot)
- **Data Processing** → Cleans and prepares data
- **AI/ML Model** → Analyzes and predicts best career options
- **Recommendation Engine** → Selects suitable careers
- **Database** → Stores career info, datasets, skills
- **Output** → Shows final career suggestions

1.Dataset Description

Before implementing the AI Career Guidance System, a dataset is required to train the machine learning model. The dataset contains information related to students and their career preferences. It plays a crucial role in helping the system learn patterns and make accurate predictions.

The dataset typically includes the following attributes:

- Academic performance (marks or CGPA)
- Stream (CSE, IT, ECE, etc.)
- Skills (programming, communication, etc.)
- Interests (AI, business, design, etc.)

- Career outcomes (Software Engineer, Data Scientist, etc.)

This dataset can be collected from surveys, educational records, or sample data created for training purposes.

3. Data Preprocessing

Before applying machine learning algorithms, the collected data is preprocessed:

- Handling missing values
- Converting categorical data into numerical format (encoding)
- Normalizing or scaling data
- Removing inconsistent or invalid inputs

This step ensures better accuracy in predictions.

4. Model Implementation

Machine learning models are used to predict suitable career options. Commonly used algorithms include:

- Decision Tree

  The model is trained using historical or sample datasets that map user profiles to career outcomes. After training, the model is tested and deployed for real-time predictions.

5. Recommendation Engine

The recommendation engine processes the output from the ML model and generates:

- Suitable career options
- Recommended courses
- Required skills for each career

It ranks the results based on relevance and accuracy.

6. System Integration

All modules (frontend, backend, ML model, and database) are integrated to work together:

- User input → Backend processing → ML model prediction → Output display

SYSTEM IMPLEMENTATION

```
from google.colab import files
uploaded = files.upload()
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder
df = pd.read_csv("dataset.csv")
print(df.head())
print(df.columns)
if len(df.columns) == 1:
    df = df[df.columns[0]].str.split(",", expand=True)
df.columns = ['Age','12th_Percentage','BTech_Stream','CGPA','Interested_Field','Career']
print(df.head())
df[['Age','12th_Percentage','CGPA']] = df[['Age','12th_Percentage','CGPA']].astype(float)
from sklearn.preprocessing import LabelEncoder
le_stream = LabelEncoder()
le_interest = LabelEncoder()
le_career = LabelEncoder()
df['BTech_Stream'] = le_stream.fit_transform(df['BTech_Stream'])
df['Interested_Field'] = le_interest.fit_transform(df['Interested_Field'])
df['Career'] = le_career.fit_transform(df['Career'])
from sklearn.tree import DecisionTreeClassifier
X = df[['Age','12th_Percentage','BTech_Stream','CGPA','Interested_Field']]
y = df['Career']
model = DecisionTreeClassifier()
```

```
model.fit(X, y)
age = int(input("Enter Age: "))
marks = float(input("Enter 12th Percentage: "))
stream = input("Enter BTech Stream (CSE/IT/ECE): ")

cgpa = float(input("Enter CGPA: "))
interest = input("Enter Interested Field: ")
stream_encoded = le_stream.transform([stream])[0]
interest_encoded = le_interest.transform([interest])[0]
prediction = model.predict([[age, marks, stream_encoded, cgpa, interest_encoded]])
result = le_career.inverse_transform(prediction)
print("🎯 Recommended Career:", result[0])
```

RESULTS AND DISCUSSION

Enter Age: 18

Enter 12th Percentage: 90

Enter BTech Stream (CSE/IT/ECE): CSE

Enter CGPA: 90

Enter Interested Field: AI

🎯 Recommended Career: AI Engineer

The AI Career Guidance System was tested using sample user input. The user entered the following details:

- Age: 18
- 12th Percentage: 90
- BTech Stream: CSE
- CGPA: 9.0

- Interested Field: AI

Based on the given input, the system successfully analyzed the data using the trained machine learning model and generated the result.

TESTING

The system was tested using sample input values.

Input:

- Age: 18
- 12th Percentage: 90
- Stream: CSE
- CGPA: 9.0
- Interest: AI

Output:

Recommended Career: AI Engineer

Result:

The system provided the correct and relevant career recommendation based on the user input.

Conclusion

The AI Career Guidance System provides a modern approach to career counseling by integrating Artificial Intelligence (AI) and Machine Learning (ML) techniques. By analyzing user data such as academic performance, skills, interests, and personal preferences, the system is able to suggest career paths that align with the individual's profile.

The system is user-friendly, efficient, and accurate, making it an effective tool for students who are confused about career choices. The testing phase shows that the system can predict appropriate career options and provide valuable insights for skill development. This reduces dependency on traditional counseling methods and enables data-driven career decision.

FUTURE ENHANCEMENTS

The system can be improved and expanded in several ways to make it more powerful and versatile:

- Larger and More Diverse Dataset

  Collecting data from multiple sources and streams to improve the prediction accuracy.

  Including real-world career success data to refine recommendations.

- Advanced AI and Deep Learning Models

  Using models like Random Forest, Gradient Boosting, or Neural Networks for better prediction.

  Implementing natural language processing (NLP) for analyzing user inputs more intelligently.

- Personality and Aptitude Testing

  Integrating personality assessment tests such as Big Five or MBTI to provide more personalized recommendations.

  Including aptitude or skill-based tests for a holistic evaluation.

- Mobile Application Development

  Developing a mobile version for easier accessibility and interaction.

  Allowing notifications for new career updates and suggestions.

- Integration with Real-Time Job Market Data

  Providing recommendations based on industry demand, salary trends, and future scope.

  Suggesting skill development courses and certifications relevant to career trends.

- Interactive User Interface and Chatbot

  Using a conversational chatbot to guide users step-by-step through the system.

  Making the system more engaging and easy to use for all age groups.

- Feedback and Continuous Learning

  Incorporating user feedback to improve the recommendation engine.

  System can learn and update recommendations automatically with new data.

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